

Clustering Techniques

Complete Study Notes — Types, Algorithms, Comparisons & Evaluation Methods

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1. What is Clustering?

DEFINITION

Clustering is an unsupervised machine learning technique that groups a set of data points such that points within the same group (cluster) are more similar to each other than to points in other groups, based on some notion of distance or similarity, without using any predefined labels.

Clustering is one of the two main tasks in unsupervised learning (alongside dimensionality reduction). Since there are no labeled outputs to learn from, the algorithm must discover structure in the data purely from the features themselves.

1.1 Why Clustering Matters

- Reveals hidden structure and natural groupings in unlabeled data
- Used as a preprocessing/exploratory step before supervised learning
- Enables segmentation, summarization, and anomaly detection
- Forms the basis for recommendation systems, market research, and biological taxonomy

1.2 Core Idea: Similarity and Distance

Almost every clustering algorithm relies on a way to quantify how "close" or "similar" two data points are. The choice of distance metric directly shapes what counts as a cluster.

Metric	Formula	Typical Use
Euclidean Distance	$\sqrt{\sum (x_i - y_i)^2}$	General-purpose, continuous numeric data
Manhattan Distance	$\sum x_i - y_i $	Grid-like data, robust to outliers
Cosine Similarity	$(x \cdot y) / (\ x\ \ y\)$	Text data, high-dimensional sparse vectors
Jaccard Distance	$1 - A \cap B / A \cup B $	Sets, binary/categorical attributes
Hamming Distance	Count of differing positions	Categorical or binary strings

EXAM TIP

A common exam question asks you to choose an appropriate distance metric for a given data type. Remember: numeric/continuous → Euclidean or Manhattan; text/sparse → Cosine; sets/binary → Jaccard or Hamming.

2. Major Families of Clustering Algorithms

Clustering algorithms can be broadly grouped into five families, each based on a different underlying assumption about what a "cluster" looks like.

1. Partitioning (Centroid-Based) Methods e.g. K-Means

Divide data into a fixed number of non-overlapping clusters, each represented by a central point (centroid or medoid). Assumes roughly spherical, similarly-sized clusters.

2. Hierarchical Methods e.g. Agglomerative

Build a nested tree of clusters (a dendrogram) either by repeatedly merging (agglomerative) or splitting (divisive) clusters. No need to fix the number of clusters upfront.

3. Density-Based Methods e.g. DBSCAN, OPTICS

Define clusters as dense regions of points separated by sparse regions. Can find arbitrarily shaped clusters and naturally identify noise/outliers.

4. Distribution-Based Methods e.g. Gaussian Mixture Models

Assume data is generated from a mixture of underlying probability distributions (usually Gaussian) and assign points probabilistically (soft clustering) to each distribution/cluster.

5. Grid-Based Methods e.g. STING, CLIQUE

Divide the data space into a finite number of grid cells, then perform clustering operations on the grid structure rather than on individual points — fast for very large, low-dimensional datasets.

BIG PICTURE

No single clustering technique works best for all datasets. The right choice depends on cluster shape, presence of noise, whether K is known, dataset size, and dimensionality.

3. Partitioning Methods

3.1 K-Means

The most widely used partitioning algorithm. It assigns each point to the nearest of k centroids, then updates each centroid as the mean of its assigned points, repeating until convergence. It minimizes the **Within-Cluster Sum of Squares (WCSS)**:

$$WCSS = \sum_{i=1}^K \sum_{x \in C_i} \|x - \mu_i\|^2$$

- 1 Choose K and initialize K centroids.
- 2 Assign each point to its nearest centroid.
- 3 Recompute each centroid as the mean of its cluster.
- 4 Repeat steps 2-3 until convergence.

3.2 K-Medoids (PAM)

Similar to K-Means, but each cluster is represented by an actual data point (a **medoid**) rather than a computed mean. This makes K-Medoids more robust to outliers and usable with any distance metric, not just Euclidean — at the cost of higher computational complexity.

Aspect	K-Means	K-Medoids
Cluster representative	Centroid (mean) — may not be a real data point	Medoid — an actual data point
Sensitivity to outliers	High — mean is pulled by extreme values	Lower — medoid is less affected by outliers
Distance metric	Restricted to Euclidean (due to mean)	Works with any distance metric
Computational cost	Lower, scales well	Higher — more expensive per iteration

EXAM TIP

If asked "why use K-Medoids over K-Means?", the answer is robustness to outliers and flexibility of distance metric — at the cost of speed.

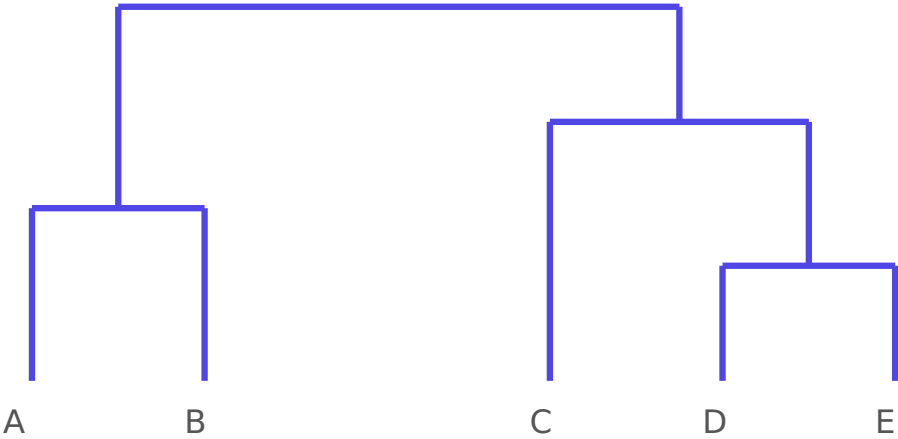
4. Hierarchical Methods

Hierarchical clustering builds a complete tree of nested clusters rather than one fixed partition, visualized using a **dendrogram**.

Approach	Strategy
Agglomerative (bottom-up)	Start with every point as its own cluster; repeatedly merge the closest pair until one cluster remains. Most common in practice.
Divisive (top-down)	Start with all points in one cluster; repeatedly split until every point is its own cluster. Computationally expensive, rarely used.

4.1 Linkage Criteria

Linkage	Rule	Effect
Single	Minimum distance between any two points across clusters	Can chain elongated clusters
Complete	Maximum distance between any two points across clusters	Compact, round clusters
Average	Mean of all pairwise distances	Balanced compromise
Ward's	Minimum increase in within-cluster variance	Similar-sized, low-variance clusters



A dendrogram: cutting at different heights yields different numbers of clusters.

EXAM TIP

Agglomerative clustering performs exactly $n - 1$ merges for n data points. Cutting a dendrogram horizontally and counting crossed lines gives the resulting number of clusters.

5. Density-Based Methods

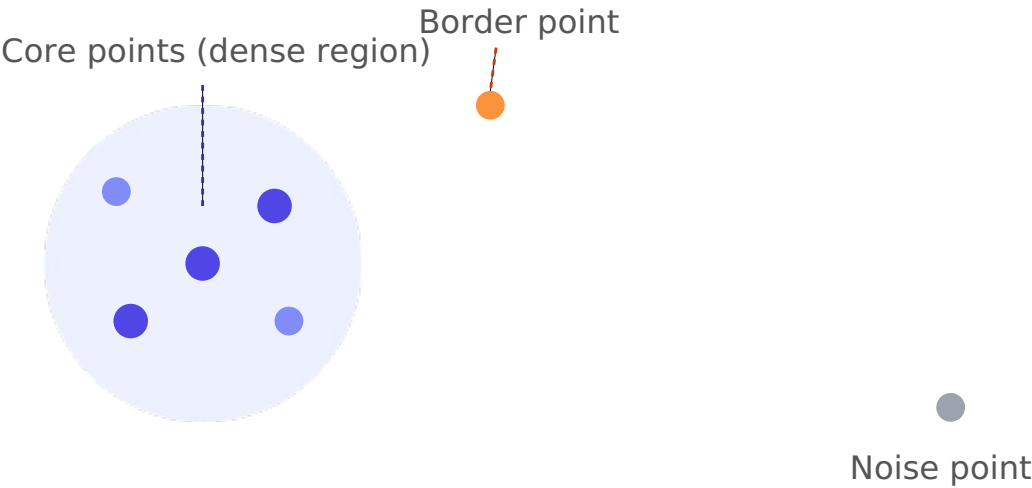
5.1 DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

DBSCAN groups together points that are closely packed, marking points in low-density regions as noise/outliers. It does not require specifying the number of clusters in advance and can discover clusters of arbitrary shape.

Parameter	Meaning
ϵ (epsilon)	Radius defining the neighborhood around a point
MinPts	Minimum number of points required within ϵ to consider a region "dense"

5.2 Point Classification in DBSCAN

- **Core point:** has at least `MinPts` points (including itself) within distance ϵ .
- **Border point:** within ϵ of a core point, but does not itself have `MinPts` neighbors.
- **Noise point:** not reachable from any core point — treated as an outlier, not assigned to any cluster.



DBSCAN classifies every point as a core, border, or noise point based on local density.

ADVANTAGES	LIMITATIONS
• No need to specify number of clusters • Finds arbitrarily shaped clusters • Automatically identifies outliers as noise	• Sensitive to choice of ϵ and <code>MinPts</code> • Struggles with clusters of very different densities • Less effective in very high-dimensional spaces

6. Distribution-Based Methods

6.1 Gaussian Mixture Models (GMM)

GMM assumes the data is generated from a mixture of several Gaussian (normal) distributions, each representing one cluster. Rather than assigning each point to exactly one cluster, GMM performs **soft clustering** — it computes the probability that each point belongs to each cluster.

$$P(x) = \sum_{i=1}^K \pi_i \cdot N(x \mid \mu_i, \Sigma_i)$$

where π_i is the mixing weight, and $N(x \mid \mu_i, \Sigma_i)$ is a Gaussian distribution with mean μ_i and covariance Σ_i .

GMM parameters are typically estimated using the **Expectation-Maximization (EM) algorithm**, which alternates between estimating cluster membership probabilities (E-step) and updating the distribution parameters (M-step).

Aspect	K-Means	GMM
Cluster assignment	Hard (each point belongs to exactly one cluster)	Soft (probability of belonging to each cluster)
Cluster shape	Spherical, equal variance assumed	Elliptical — can model varying size/orientation
Underlying model	Distance-based heuristic	Probabilistic generative model

EXAM TIP

GMM can be viewed as a probabilistic generalization of K-Means — K-Means is roughly equivalent to GMM with equal, spherical covariance and hard assignment.

7. Grid-Based Methods (Brief Overview)

Grid-based methods divide the data space into a finite number of cells forming a grid structure, then perform all clustering operations on this grid rather than on individual data points. Examples include **STING** (Statistical Information Grid) and **CLIQUE** (Clustering In QUEst).

KEY ADVANTAGE

Because processing time depends on the number of grid cells rather than the number of data points, grid-based methods are very fast and scale well — particularly useful for large, low-dimensional spatial datasets.

8. Comparing the Major Techniques

Technique	Needs K?	Cluster Shape	Handles Noise?	Scalability
K-Means	Yes	Spherical, similar size	Poor	High
K-Medoids	Yes	Spherical, similar size	Moderate	Moderate
Hierarchical (Agglomerative)	No (choose after dendrogram)	Depends on linkage	Poor-Moderate	Low ($O(n^2)+$)
DBSCAN	No	Arbitrary	Excellent (built-in)	Moderate
Gaussian Mixture Models	Yes (number of components)	Elliptical	Poor	Moderate
Grid-Based (STING, CLIQUE)	Varies	Grid-aligned regions	Moderate	Very High

8.1 How to Choose a Technique

- **Know K and expect round, similar-sized clusters?** → K-Means
- **Need a full hierarchy / nested structure, or unsure of K?** → Hierarchical Clustering
- **Clusters are irregularly shaped, with noise/outliers present?** → DBSCAN
- **Clusters overlap, or you need probability of membership?** → Gaussian Mixture Model
- **Data has many outliers and you need a robust centroid?** → K-Medoids
- **Very large, low-dimensional spatial dataset?** → Grid-based methods

9. Evaluating Clustering Results

Because clustering is unsupervised, evaluation is trickier than for supervised learning. Metrics fall into two categories: **internal** (no ground truth needed) and **external** (compared against known labels).

9.1 Internal Evaluation Metrics

Metric	What It Measures
WCSS / Inertia	Total squared distance of points to their cluster centroid — lower is more compact
Silhouette Score	How well each point fits its own cluster vs the nearest other cluster, ranges from -1 to $+1$
Davies-Bouldin Index	Ratio of within-cluster scatter to between-cluster separation — lower is better
Dunn Index	Ratio of minimum inter-cluster distance to maximum intra-cluster distance — higher is better

9.2 External Evaluation Metrics

Used only when true class labels are known (e.g. for benchmarking on a labeled dataset):

Metric	What It Measures
Adjusted Rand Index (ARI)	Similarity between predicted clusters and true labels, adjusted for chance
Normalized Mutual Information (NMI)	Shared information between predicted clusters and true labels
Fowlkes-Mallows Index	Geometric mean of pairwise precision and recall between clusters and true labels

EXAM TIP

If a question gives you ground-truth labels, use external metrics (ARI, NMI). If no labels exist (the typical real-world case), use internal metrics (Silhouette, Davies-Bouldin).

10. Real-World Applications

- **Customer Segmentation:** grouping customers by behavior for targeted marketing.
- **Anomaly/Fraud Detection:** flagging points that don't fit into any dense cluster (e.g. via DBSCAN).
- **Image Segmentation & Compression:** grouping similar pixels or colors.
- **Bioinformatics:** phylogenetic trees, gene expression grouping.
- **Document/Topic Clustering:** organizing articles, search results, or social media posts.
- **Geospatial Analysis:** identifying regional hotspots, e.g. crime clusters or delivery zones.

11. Summary & Quick Revision

Key Takeaways

- Clustering groups unlabeled data based on similarity/distance, with no predefined classes.
- Five major families: **Partitioning** (K-Means, K-Medoids), **Hierarchical** (Agglomerative/Divisive), **Density-based** (DBSCAN), **Distribution-based** (GMM), and **Grid-based** (STING, CLIQUE).
- K-Means/K-Medoids need K specified and assume spherical clusters; Hierarchical needs no K but is computationally heavier; DBSCAN finds arbitrary shapes and handles noise natively; GMM gives soft, probabilistic cluster membership.
- Distance metric choice (Euclidean, Manhattan, Cosine, Jaccard, Hamming) should match the data type.
- Evaluate clustering using internal metrics (Silhouette, WCSS, Davies-Bouldin) when no labels exist, or external metrics (ARI, NMI) when ground truth is available.
- Technique choice depends on whether K is known, cluster shape, presence of noise, and dataset size/dimensionality.

11.1 Practice Questions

Q1. Which clustering family is best suited for discovering clusters of arbitrary shape with embedded noise?

A1. Density-based methods, such as DBSCAN, since they define clusters by dense regions and explicitly label sparse points as noise.

Q2. What is the key structural difference between K-Means and Gaussian Mixture Models?

A2. K-Means performs hard assignment (each point belongs to exactly one cluster) and assumes spherical clusters, while GMM performs soft, probabilistic assignment and can model elliptical clusters of varying shape and orientation.

Q3. Why might K-Medoids be preferred over K-Means for a dataset with significant outliers?

A3. Because K-Medoids uses an actual data point (medoid) as the cluster representative rather than a mean, making it less sensitive to extreme outlier values that would otherwise distort a centroid.

Q4. What do the parameters ϵ and MinPts control in DBSCAN?

A4. ϵ defines the radius of the neighborhood around a point, and MinPts defines the minimum number of points required within that radius for the region to be considered dense (i.e. for the point to be a core point).

Q5. When should you use external evaluation metrics like ARI instead of internal metrics like Silhouette Score?

A5. When ground-truth class labels are available for comparison, such as during benchmarking on a labeled dataset; otherwise, internal metrics are used since no true labels exist in typical unsupervised settings.

Q6. Why are grid-based methods especially fast for large datasets?

A6. Because their computational cost depends on the number of grid cells rather than the number of individual data points, making them scale well even as the dataset size grows.